Progressive exercises

E1 · Practice. Consider two events A and B such that $P(A) = 0.3$, $P_A(B) = 0.8$ and $P_{\overline{A}}(B) = 0.1$. Calculate $P(B)$, then $P_B(A)$.

E2 · Practice. The random variable X follows the binomial distribution with parameters 8 and $\frac{1}{4}$. Give the exact values of $P(X = 3)$, $E(X)$ and $V(X)$.

E3 · Practice. Let X be a random variable taking integer values. For every integer k, set $F(k) = P(X \leq k)$. Express $P(X \geq 4)$ and $P(2 \leq X \leq 5)$ in terms of F.

E4 · Practice. Five cards are drawn in succession, without replacement, from a pack of 52 cards containing four aces. Let X be the number of aces obtained. Does the variable X follow a binomial distribution? Justify.

E5 · Practice. A trial whose probability of success is $\frac{1}{2}$ is repeated independently. Determine the smallest number of trials for which the probability of obtaining at least one success is greater than or equal to 0.99.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 1, question 5 (p. 3). In class after S41. Read the context of questions 1 to 4; for this work, accept the result $p = 0.0714$ from question 4. The paper is available here: official past paper. Use the data and instructions of this past paper.